

Factor	Apache Iceberg	Apache Hudi	Delta Lake
ACID transaction	Present	Present	Present
Partition evolution	Present	Absent	Absent
Schema evolution	Present	Partial	Limited
Time-travel	Present	Present	Present
Project governance	Apache projects with a diverse PMC (top level project)	Apache projects with a diverse PMC (top level project)	Linux foundation projects with all data bricks TSC
Tool read compatibility	Apache Hive, Demio Sonar, Apache Flink, Apache Spark, Presto, Trino, Athena, Snowflake, Databricks Spark, Apache Impact, Apache Drill	Apache Hive, Demio Sonar, Apache Flink, Apache Spark, Presto, Trino, Athena, Databricks Spark, Apache Impact, Red Shift, Big Query	Apache Hive, Demio Sonar, Apache Flink, Databricks SQL, Analytics, Presto, Trino, Athena, Snowflake, Red Shift, Apache Bea
Tool write compatibility	Apache Hive, Demio Sonar, Apache Flink, Apache Spark, Presto, Trino, Athena, Snowflake, Databricks Spark, Debezium	Apache Flink, Apache Spark, Databricks Spark, Debezium, Kafka Connect	OSS Delta Lake, Apache Spark, Trino, Apache Flink, Databricks Spark, Kafka Connect, Debezium, Distributed Delta Lake

Table 1: A comparison of Apache Iceberg, Apache Hudi and Delta Lake (Table credit: *dremio.com*)

One of the important features of Iceberg is partitioning, a technique used for optimising data lakes. For example, for sales data where queries often rely on when the sales occurred, you may partition data by the day, month, or year of the sale.

Data lakes aim to democratise data. However, it is necessary to reduce the complexity of data structures and physical data storage. In 2009, Facebook released a Hive table format that could not answer this problem. In contrast, Iceberg, however, gives table format specifications with a set of APIs and libraries to interact with tables.

Unless Apache Hive adds it, removing and updating data is very fast and easy in Iceberg. It supports SQL commands. So, it can be used to work with Hive, Spark, Impala, and so on. Such table formats are becoming the format of choice for large data sets (of sizes ranging from hundreds of terabytes to hundreds of petabytes) across the data industry.

Over the years, since Netflix donated Iceberg to Apache, it has been adopted by several major tech companies, including Adobe, Airbnb, Apple, Google, LinkedIn, Snapshot and Snowflake.

Apache Iceberg and AI

As more organisations are navigating the uncharted sea called 'Enterprise AI', data management expert Cloudera is now adopting Apache Iceberg.

With the open data lakes created by Apache Iceberg, customers will get strong analytics and AI capabilities

for their enterprise data, irrespective of whether the data is cloud based or on-premise. It is becoming a key technology for multi-vendor data ecosystems to get the most from AI. Cloudera recently claimed that its Iceberg-based solutions are managing 25 million terabytes of data.

With the increased support of Apache Iceberg, generative AI and large language models (LLMs), organisations can utilise data effectively. This further enables users to tap into more of their data and leverage it in different ways. **END** 

References

- Article 'Why Apache Iceberg will rule data in the cloud' by James Malone
- Blog 'Apache Iceberg: An architectural look under covers'; *dremio.com*
- Article 'Apache Iceberg becoming large scale analytics of choices' by David Curry
- Certain articles appeared in *Computer Weekly*

By: Vinayak R. Adkoli

The author holds a B.E. in industrial production and has served as a lecturer in three different polytechnics for 10 years. He is also a freelance writer and cartoonist.